NECESSARY CONDITIONS FOR PHOTOSYNTHESIS

1. Carbon dioxide
2. Water
3. Chlorophyll
4. Sunlight

CARBON DIOXIDE

- ✓ Plants get their carbon dioxide from the air.
- ✓ This gas comes from animals (e.g. people) when they respire.

- ✓ It can also be produced from other chemical reactions.
- ✓ Chloroplasts uses carbon dioxide as they manufacture food.
- ✓ This makes the concentration of carbon dioxide inside the cells lower than the air surrounding the leaf.
- ✓ This create concentration gradient > Difference in concentration between two regions.
- ✓ As a result carbon dioxide move into the leaf through stomata by diffusion.

WATER

- ✓ Plants get their water from the soil.
- ✓ Water is taken up (absorbed) by root hairs and move across the root into the xylem vessels.
- ✓ Then it flows up through the xylem vessels which passes through the center of the root into the stem.
- ✓ Finally the water gets into the leaves.

SUNLIGHT

- ✓ Plants get light from the sun.
- ✓ Sunlight usually falls into the upper surface of leaves.
- ✓ It passes through epidermis into the mesophyll.
- ✓ Since epidermis cells have no chloroplasts they allow light to pass through them easily.
- ✓ The palisade cells are nearer to the top of the leaf than the spongy cells.
- ✓ Therefore the palisade cells get more sunlight and it is where photosynthesis happens most.

CHROLOPHYLL

- ✓ Plants have chloroplasts which contains chlorophyll.

Types of pigments in leaves

A pigment is any coloured substance found in cells. Leaves have three types of pigments.

Plant pigments are mostly found in the chloroplast of the cells. They include;

(i) Chlorophyll

(ii) Carotene

(iii) Xanthophylls

To carry out this study, we will first carry out the following activity.

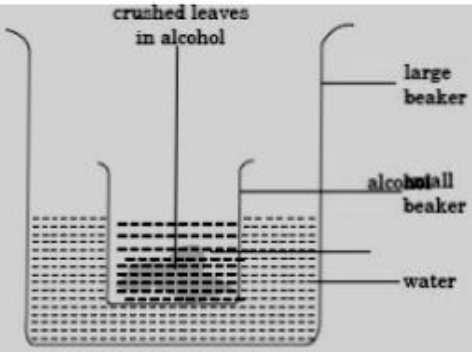
Activity 3.3: To investigate types of pigments in plant leaves

Requirements

- Green leaves
- Alcohol
- Filter paper
- Tape
- Beakers – small and large.
- Mortar and pestle
- Glass rod

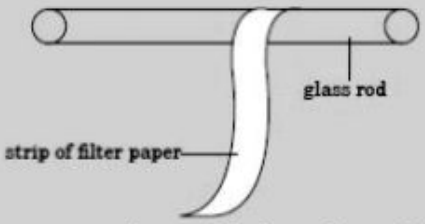
Procedure

1. Collect green leaves from plants in the school compound.
2. Chop the leaves into small pieces or crush them using pestle and mortar.
3. Place the crushed leaves into a small glass beaker.
4. Add alcohol to half fill the beaker.
5. Place the beaker into a large glass beaker half filled with water as shown below and then heat the large beaker. The large beaker with water is called a water bath.
6. Heat the beaker until the alcohol turns green. This shows that pigments in the leaves have dissolved into the alcohol.



Note: Do not heat alcohol directly because it catches fire easily

7. Remove the alcohol mixture from the beaker. Filter it into another small beaker
8. Cut the small long strip of filter paper about 3 cm wide and 10 cm long. Stick it unto a glass rod using a sellotape as shown below.



9. Suspend the filter paper strip into a container with the coloured alcohol such that the tip of the paper touches the mixture.
10. Allow the strip to absorb the coloured alcohol for about 30 minutes. Observe the colours formed on the strip from the bottom to the top.

Discussion

From the activity, you may have realised that the filter paper strip started to absorb the coloured alcohol. The absorbed substance rose up the paper forming different shades of colours.

The first and prominent colour to appear was green. Then three other colours appeared that included light green, yellow and light yellow.

The green colour represents the chlorophyll pigment. The yellow colour represents carotene pigment and the light yellow colour represents xanthophyll pigment.

1. Chlorophyll

This is the main pigment found in plant leaves. It is green in colour.

2. Carotene

Carotene is a pigment that is usually yellow, orange or red in colour.

3. Xanthophyll

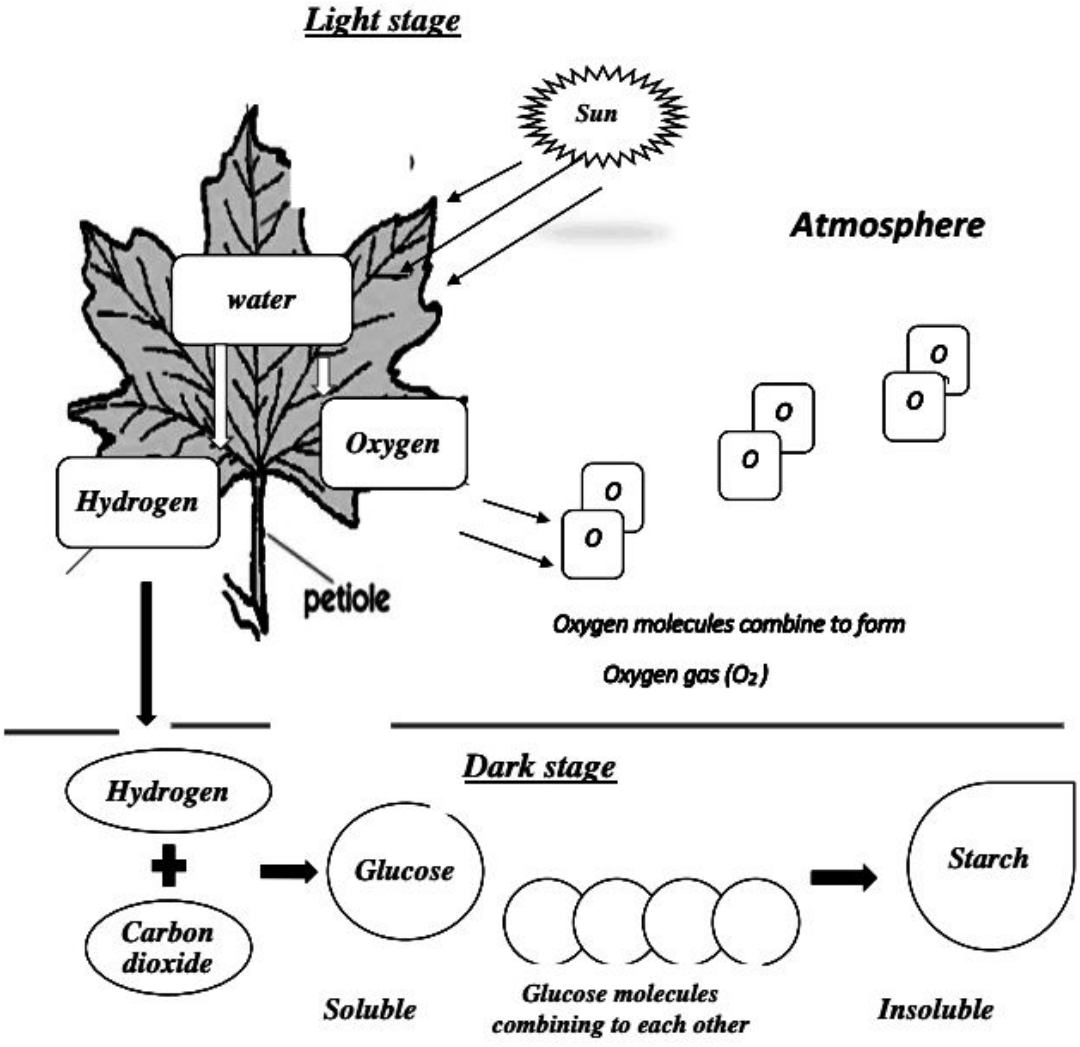
This is a pigment that is usually light yellow in colour. It absorbs lesser amount of green and blue light.

LIGHT STAGE

- The chlorophyll in plants capture sunlight and convert it into chemical energy.
- This chemical energy (adenosine triphosphate (ATP)) break down **water** into **hydrogen** and **oxygen molecules**.
- Oxygen molecules arrange themselves to form **oxygen gas.**
- Some of oxygen gas remain in the plant for respiration. However a lot of oxygen gas is released into the atmosphere since is a **by-product**.
- Therefore **water is a raw material** in light stage and **oxygen is a product in light stage**.
- **Hydrogen molecules** are stored by the plant because they are used in dark stage for photosynthesis.

DARK STAGE

- **Carbon dioxide** combine (react) with **hydrogen gas** produced in the light stage to produce **glucose**.
- Glucose is **soluble in water** therefore a plant cannot store it in this form.
- It is converted to **starch** which is **insoluble in water**. Glucose molecules combine to form starch for future use.
- Therefore **carbon dioxide is raw material** in dark stage and **glucose is the product** in dark stage.



PROCESS DURING PHOTOSYNTHESIS

DIFFUSION is the movement of molecules from a region of high concentration to a region of low concentration.

- ✓ CO_2 moves into the leaf from the atmosphere by diffusion.
- ✓ O_2 moves out of leaf into the atmosphere by diffusion.

OSMOSIS is the net movement of water molecules from a high concreted region to a lower concentrated region.

- ✓ Water moves from the soil into the plant by osmosis.

PHOTOPHOSPHORATION is the process whereby chlorophyll absorb sunlight and convert it to chemical energy.

PHOTOLYSIS is the breakdown of water molecules into hydrogen and oxygen molecules by sunlight energy

CONDENSATION is the process which involves the combination of glucose molecules to produce starch. A large molecule is formed from small molecules.

REDUCTION is the breakdown of starch with aid of oxygen gas to produce glucose for immediate use.

ADAPTATIONS OF LEAVES FOR PHOTOSYNTHESIS

1. Have chloroplasts > contains chlorophyll that capture sunlight.
2. Have broad, thin and flat lamina > to absorb more sunlight.
3. Have a petiole or leaf stalk >it expose the leaf to sunlight and attach the leaf to the stem for support.
4. Has stomata > for gaseous exchange.
5. Have a network of veins > to supply water to the leaf.
6. Have air spaces in spongy mesophyll > to allow carbon dioxide and oxygen to diffuse in and out of cells respectively.
7. Have transparent epidermis > to allow sunlight to penetrate into the cells

The fate of glucose after photosynthesis

As we mentioned earlier ***glucose*** is produced as the product during the dark stage. Glucose is the basic organic food substance. After photosynthesis, glucose is utilised in the following ways.

1. Respiration

Some of the glucose is used by the plant cells for respiration to release energy.

2. Storage

Most of the glucose produced in the leaf is usually converted to starch. Starch is stored in plant tissues. Some glucose is combined with fructose to form sucrose. Sucrose is stored in stems such as the stems of sugarcanes.

3. Formation of cellulose

Some glucose is converted into a complex substance called *cellulose.* Cellulose is used in making of cell walls.

4. Converted into lipids

Lipids are stored in plant tissues like the seed endosperm.

5. Glucose is combined with nitrates to form proteins